

A domination proof that $\mathcal{W}^2(K) = \widetilde{\mathcal{W}}^2(K)$ for the two-hidden-layer Barron-tree question

FA/Banach run `fa_banach_001`, lane 1

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Source Metadata

- Source paper: Weinan E and Stephan Wojtowytsch, *On the Banach spaces associated with multi-layer ReLU networks: Function representation, approximation theory and gradient descent dynamics*, arXiv:2007.15623.
- Local source PDF: `source_paper.pdf`.
- Extracted question location: Section 4.3, page 26 of the arXiv PDF.
- Claimed status: likely full solution of the two-hidden-layer equality question $\mathcal{W}^2(K) = \widetilde{\mathcal{W}}^2(K)$.

Open Problem Extract

The source paper defines the neural tree space $\mathcal{W}^2(K)$ and the network-like two-hidden-layer space

$$X_{(0,1),(0,1),\{0,\dots,d\};K} = \bigcup_{\bar{\pi}} \mathcal{B}_{Y_{\bar{\pi}},K} =: \widetilde{\mathcal{W}}^2(K),$$

where $Y_{\bar{\pi}}$ denotes the Barron functions representable by an $L^1(\bar{\pi})$ density in the first hidden layer. After observing that no single $\bar{\pi}$ can represent every Barron function when K is the closure of an open set, the paper asks whether this loss of first-layer nonseparability actually reduces expressivity:

“The question whether this reduces expressivity or whether in fact $\mathcal{W}^2 = \widetilde{\mathcal{W}}^2$ remains open.”

may be dense in $C^0(K)$. If $\bar{\pi}$ is the uniform distribution on S^d , then $Y_{\bar{\pi}}$ is dense in C^0 since $L^1(\bar{\pi})$ is dense in the space of Radon measures on S^d with respect to the weak topology.

Claim: There is no distribution $\bar{\pi}$ on S^d such that every Barron function can be expressed with an L^1 -density with respect to $\bar{\pi}$ if K is the closure of an open set.

Proof of claim: Barron space is not separable since

$$\|\sigma(w_1^T \cdot) - \sigma(w_2^T \cdot)\|_{B^1(K)} \geq [\sigma(w_1^T \cdot) - \sigma(w_2^T \cdot)]_{C^{0,1}(K)} \geq 1$$

if one of the hyperplanes $\{x : w_{1/2}^T x = 0\}$ intersects the interior of K . This is the case for uncountably many $w \in S^d$. On the other hand, $L^1(\bar{\pi})$ (and also its quotient by the kernel of the realization map) is separable for any Radon measure. Thus the two spaces cannot coincide. $\square$

The claim can be phrased and proved in greater generality if K is a manifold or similar. We note that for fixed $\bar{\pi}$, the space $Y_{\bar{\pi}}$ embeds continuously into $C^{0,1}(K)$, but its unit ball is not closed in $C^0(K)$. Nevertheless, we may consider the space

$$\mathcal{B}_{Y_{\bar{\pi}}, K} = \left\{ f_{\mu}(x) = \int_{B^X \cap Y_{\bar{\pi}}, K} \sigma(g(x)) \mu(dg) \mid \mu \text{ admissible} \right\}$$

where admissible measures are finite (signed) Radon measures for which $Y_{\bar{\pi}}$ is measurable. Every distribution $\bar{\pi}$ on S^d can be obtained as the push-forward of Lebesgue measure on the unit interval along a measurable map $\phi : (0, 1) \rightarrow S^d$, see e.g. [EW20b, Section 2.8]. Thus the associated space of neural networks with two hidden layers is

$$X_{(0,1),(0,1),\{0,\dots,d\};K} = \bigcup_{\bar{\pi}} \mathcal{B}_{Y_{\bar{\pi}}, K} =: \widehat{\mathcal{W}}^2(K)$$

where the union is over all probability distributions $\bar{\pi}$ on S^d . Thus the first layer of $f \in \mathcal{W}^2$ is wide enough to contain the entire unit ball of $\mathcal{W}^1$, while the first layer of $f \in \widehat{\mathcal{W}}^2$ can only express a separable subset of the unit ball in $\mathcal{W}^1$. The question whether this reduces expressivity or whether in fact $\mathcal{W}^2 = \widehat{\mathcal{W}}^2$ remains open.

Finally, consider the space $\mathcal{W}_{\mathcal{L}^1, \mathcal{L}^1, \pi^0}(K)$ where the weights of a function satisfy

$$a^2 \in L^2(0, 1), \quad a^1 \in L^2((0, 1)^2), \quad a^0 \in L^2((0, 1) \times \{0, \dots, d\}) = L^2((0, 1); \mathbb{R}^d).$$

We proceed as before, but normalize with respect to L^2 rather than L^1/L^∞ . Again, we can consider the maps

$$\Psi : (0, 1) \times (0, 1) \rightarrow \mathbb{R}^{d+2}, \quad (\theta_2, \theta_1) \mapsto (a_{\theta_2 \theta_1}^1, a_{\theta_1}^0)$$

and note as before that

$$\int_0^1 a_{\theta_2, \theta_1}^1 \sigma \left(\sum_{\theta_0=1}^{d+1} a_{\theta_1, \theta_0}^0 x_{\theta_0} \right) d\theta_1 = \int_{\mathbb{R}^{d+1}} \bar{a}^{\theta_2}(w) \sigma(w^T x) \bar{\pi}(dw)$$

where this time $\bar{a} \in L^2(\bar{\pi})$ for almost all $\theta_2 \in (0, 1)$. Thus the first layer of $f \in \mathcal{W}_{\mathcal{L}^1, \mathcal{L}^1, \pi^0}$ takes values in a single reproducing kernel Hilbert space $\mathcal{H}_{\bar{\pi}}$ associated to the kernel

$$k_{\bar{\pi}}(x, x') = \int_{\mathbb{R}^{d+1}} \sigma(w^T x) \sigma(w^T x') \bar{\pi}(dw)$$

while the first layer of $f \in \mathcal{W}^2$ may be wide enough to contain every function in the unit ball of Barron space. Again, the relationship between the function spaces remains open.

Proof Intuition

The nonseparability obstruction in the paper rules out one universal first-layer distribution $\bar{\pi}$ that works for every Barron function. But a single element of $\mathcal{W}^2(K)$ does not need every Barron function. It is represented by one finite outer Radon measure μ on the Barron unit ball.

For μ -almost every inner Barron function g , choose one first-layer representing measure ν_g of norm at most one. Average the total variations $|\nu_g|$ against the outer variation measure $|\mu|$. This produces a single probability measure $\bar{\pi}$ that dominates ν_g for $|\mu|$ -almost every g . Hence almost every inner g used by this particular $\mathcal{W}^2$ representation lies in the separable subspace $Y_{\bar{\pi}}$. The same outer integral is then a $\widetilde{\mathcal{W}}^2$ representation.

Thus the first-layer distribution may depend on the target function, and that flexibility is enough to remove the apparent expressivity gap.

Theorem

Theorem 1. *Let $K \subset \mathbb{R}^d$ be compact. With the notation of E-Wojtowytsch, for two hidden layers,*

$$\mathcal{W}^2(K) = \widetilde{\mathcal{W}}^2(K) = X_{(0,1),(0,1),\{0,\dots,d\};K}.$$

In particular, the network-like L^1 indexed model with two hidden layers has the same function expressivity as the two-hidden-layer neural tree space.

Proof

The inclusion $\widetilde{\mathcal{W}}^2(K) \subseteq \mathcal{W}^2(K)$ is already established in the source paper: every two-hidden-layer indexed network gives a $\mathcal{W}^2$ representation whose outer measure is concentrated on some $Y_{\bar{\pi}}$.

We prove the reverse inclusion. Let P be the compact first-layer parameter space used to represent the Barron space $\mathcal{W}^1(K)$; equivalently, after the ReLU normalization in the source paper, one may take $P = S^d$. Let

$$T : \mathcal{M}(P) \longrightarrow C(K), \quad T\nu(x) = \int_P \sigma(p(x)) d\nu(p)$$

be the realization map. The closed unit ball B of $\mathcal{W}^1(K)$ is the image under T of the total-variation unit ball

$$\mathcal{M}_1(P) := \{\nu \in \mathcal{M}(P) : \|\nu\|_{\text{TV}} \leq 1\}.$$

Indeed, the definition gives the image inclusion, and compactness of $\mathcal{M}_1(P)$ in the weak-star topology gives norm-attainment for points of B . The map T is continuous from $\mathcal{M}_1(P)$ with its weak-star topology into $C(K)$.

Let $f \in \mathcal{W}^2(K)$. Choose a finite signed Radon measure μ on B such that

$$f(x) = \int_B \sigma(g(x)) d\mu(g).$$

If $\mu = 0$, there is nothing to prove. Otherwise write $\lambda = |\mu|$.

For each $g \in B$, consider the compact nonempty fiber

$$\Gamma(g) := \{\nu \in \mathcal{M}_1(P) : T\nu = g\}.$$

The graph

$$\{(g, \nu) \in B \times \mathcal{M}_1(P) : T\nu = g\}$$

is closed between compact metrizable spaces. By the standard measurable selection theorem for compact-valued Borel multifunctions, there exists a Borel map $g \mapsto \nu_g \in \mathcal{M}_1(P)$ such that $T\nu_g = g$ for all $g \in B$.

Define a finite positive measure ρ on P by

$$\rho(A) = \int_B |\nu_g|(A) d\lambda(g)$$

for Borel $A \subseteq P$, and set $\bar{\pi} = \rho/\rho(P)$ if $\rho(P) > 0$; if $\rho(P) = 0$, choose any probability measure on P . Since $\|\nu_g\|_{\text{TV}} \leq 1$, the measure ρ is finite.

If $\bar{\pi}(A) = 0$, then $\rho(A) = 0$, so $|\nu_g|(A) = 0$ for λ -almost every g . Hence

$$\nu_g \ll \bar{\pi} \quad \text{for } \lambda\text{-almost every } g.$$

For such g , the Radon–Nikodym theorem gives $h_g \in L^1(\bar{\pi})$ with

$$d\nu_g = h_g d\bar{\pi}, \quad \|h_g\|_{L^1(\bar{\pi})} = \|\nu_g\|_{\text{TV}} \leq 1.$$

Therefore $g = T\nu_g$ belongs to the subspace $Y_{\bar{\pi}}$ of Barron functions representable by an $L^1(\bar{\pi})$ density.

Thus $\lambda(B \setminus Y_{\bar{\pi}}) = 0$. Since μ is absolutely continuous with respect to λ , the original outer integral is, after changing nothing on a μ -null set, an admissible finite signed Radon measure concentrated on $B \cap Y_{\bar{\pi}}$:

$$f(x) = \int_{B \cap Y_{\bar{\pi}}} \sigma(g(x)) d\mu(g).$$

This is precisely a representation of f as an element of $\mathcal{B}_{Y_{\bar{\pi}}, K} \subseteq \widetilde{\mathcal{W}}^2(K)$.

Consequently $\mathcal{W}^2(K) \subseteq \widetilde{\mathcal{W}}^2(K)$. Together with the known reverse inclusion, this proves

$$\mathcal{W}^2(K) = \widetilde{\mathcal{W}}^2(K).$$

Finally, the source paper identifies $\widetilde{\mathcal{W}}^2(K)$ with $X_{(0,1),(0,1),\{0,\dots,d\};K}$ by pushing any probability $\bar{\pi}$ forward from Lebesgue measure on $(0,1)$, so the displayed equality follows. $\square$

Measurability Note

The source paper allows admissible outer measures for which $Y_{\bar{\pi}}$ is measurable. This condition is satisfied here. The set $Y_{\bar{\pi}}$ is the realization image of the Polish space $L^1(\bar{\pi})$ under the continuous map

$$h \mapsto \left[x \mapsto \int_P h(p) \sigma(p(x)) d\bar{\pi}(p) \right],$$

and hence is analytic in the compact metric space $B \subset C(K)$. Analytic sets are universally measurable, so $Y_{\bar{\pi}}$ is measurable for the finite Radon measure μ used above.

Verification Notes

- The proof does not use the source paper’s extra assumption that K is the closure of an open set; compactness of K suffices.
- The proof answers the equality question $\mathcal{W}^2 = \widetilde{\mathcal{W}}^2$ for the L^1 indexed/two-hidden-layer space.
- The proof does not settle the separate question about the Hilbert-weight/RKHS subspace $\mathcal{W}_{\mathcal{L}^1, \mathcal{L}^1, \pi^0}(K)$ versus $\mathcal{W}^2(K)$.
- The only external tools are standard compact measurable selection and Radon–Nikodym domination.

Novelty Check

Local run indexes were searched for arXiv id 2007.15623, the title, “neural tree spaces”, “Barron”, “Hilbert weights”, and “ $\widetilde{\mathcal{W}}^2$ ” variants; no duplicate solution or attempt for this problem was found.

On 2026-07-03, bounded web searches were run for exact and close phrases including “W2 equals tilde W2 Barron”, “neural tree spaces tilde W2”, “arbitrarily wide neural networks Hilbert weights”, and the source title with “remains open”. The search surfaced the source arXiv page but no later explicit resolution of this equality question. Novelty confidence is therefore moderate-to-high, subject to specialist review in the neural-network approximation literature.

Human Review Recommendation

Review as a likely full solution of the source paper’s two-hidden-layer expressivity question. The main point to check is the measurable-selection/Radon–Nikodym step and whether the source paper’s identification

$$X_{(0,1),(0,1),\{0,\dots,d\};K} = \bigcup_{\pi} \mathcal{B}_{Y_{\pi},K}$$

is intended with exactly the admissibility convention used here.

References

- [1] Weinan E and Stephan Wojtowytsch, *On the Banach spaces associated with multi-layer ReLU networks: Function representation, approximation theory and gradient descent dynamics*, arXiv:2007.15623, 2020.
- [2] Alexander S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.
- [3] Gerald B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999.